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B.TECH/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2025

Fifth & Sixth Semester

21CSC303J - SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks) — Answer **all** Questions

		Marks	BL	CO
1.	If you were a lead developer of a software company and you are asked to submit a project/product within a stipulated timeframe with no cost barriers, which model would you select? (A) Waterfall (B) Spiral (C) RAD (D) Incremental	1	1	1
2.	Identify the one that given below is does not account for software failure. (A) Growing Demand (B) Low anticipation (C) Expanding Supply (D) Less reliable and expensive	1	1	1
3.	The spiral model was first described by (A) IBM (B) Barry Boehm (C) Pressman (D) Royce	1	1	1
4.	Which of the following activity is not timeboxed? (A) Sprint Retrospective (B) Sprint (C) Product Backlog Refinement (D) Daily Scrum	1	1	1
5.	"Consider a smart home system where, a heat sensor detects an intrusion and alerts the security company." What kind of a requirement the system is providing? (A) Functional (B) Non-Functional (C) Known Requirement (D) Software requirement	1	2	2
6.	Identify the kind of approach was introduced for elicitation and modelling to give a functional view of the system? (A) Object Oriented Design (B) Use Cases (C) Fusion (D) Object Modeling Technique	1	1	2
7.	Which document typically contains user requirements? (A) Software Design Document (SDD) (B) User Manual (C) Software Requirements Specification (SRS) (D) Test Plan	1	1	2
8.	System requirements are generally written for (A) End-users (B) Developers and testers (C) Marketing teams (D) Legal departments	1	1	2
9.	How does the term "design model" refer to in software engineering? (A) A model that provides a detailed blueprint of the software (B) The final implementation of the software (C) The hardware configuration needed for the software (D) A set of algorithms used in the software	1	1	3
10.	What is the purpose of architectural design in software engineering?			

(A) To determine the software's visual appearance	(B) To define the overall structure of the system	1	1	3
(C) To write the program code	(D) To test the software components			
11. Choose the correct option, "This classic program structure decomposes function into a control hierarchy where a main program invokes a number of program components that in turn may invoke still other components".		1	1	3
(A) Data Flow Architecture	(B) Call and Return Architecture			
(C) Object-oriented Architecture	(D) Layered Architecture			
12. Identify the one element of a modern GUI typically provides navigation between different views or windows?		1	1	3
(A) Menu bar	(B) Scroll bar			
(C) Icon	(D) Error-message			
13. Which of the following is the primary purpose of a review in software development?		1	1	4
(A) Identifying bugs	(B) Improving team collaboration			
(C) Ensuring code quality and verifying requirements	(D) Providing technical training			
14. What is the purpose of a desk check in software development?		1	1	4
(A) To formally test the code using automated tools	(B) To manually review code by walking through it line by line			
(C) To create a checklist for code standards	(D) To ensure deployment readiness			
15. What is black-box testing?		1	1	4
(A) Testing with no access to code	(B) Testing with full code access			
(C) Testing only for security flaws	(D) Testing only the database			
16. Which testing technique uses knowledge of code structure?		1	1	4
(A) Black-box testing	(B) White-box testing			
(C) Functional testing	(D) System testing			
17. Identify the one type of software testing is generally used in Software Maintenance.		1	1	5
(A) Regression Testing	(B) System Testing			
(C) Integration Testing	(D) Unit Testing			
18. How much effort is typically expended by a software organization on software maintenance?		1	1	5
(A) 20 percent	(B) 40 percent			
(C) 60 percent	(D) 80 percent			
19. The modification of the software to match changes in the ever-changing environment, falls under which category of software maintenance?		1	1	5
(A) Corrective	(B) Adaptive			
(C) Perfective	(D) Preventive			
20. What is the first step in risk management?		1	1	5
(A) Risk mitigation	(B) Risk identification			
(C) Risk control	(D) Risk transfer			

PART - B (5 × 8 = 40 Marks) — **Marks BL CO**

Answer **all** Questions

21. (a) Given Covide-19 in early 2019, you have been asked to support as an IT person to an APP as quickly as possible before the situation gets worsen and disastrous, drugs and other crucial essentials between patients, hospitals and pharmacy. Propose which traditional model would you adopt, why did you adopt this, what are the advantages, do identity the stake holders. Also, list if there are any challenges that exist in this model as well.

(OR)

	(b)	Assume that you are a software developer of a company. A client has approached you for a better solution to the problem faced by their side. The client stated that risks/uncertainties will lead to loss if not properly planned and solve. Justify with a next pictorial representation, which model will you opt for software development and mention its merit and demerit in detail.	8	4	1
22.	(a)	You are a WebApp designer for future learning corporation, a distance learning company. You intend to implement an internet-based "learning engine" that will enable you to deliver course content to a student. The learning engine provides the basic infrastructure for delivering learning content on any subject (content designer will prepare appropriate content). Develop a prototype interface design for the learning engine.	8	3	2
		(OR)			
	(b)	For a given project was estimated with a size of 600 KLOC. Calculate the effort, scheduled time for development. Also, calculate the average staff size and productivity of the software for organic, semidetached, embedded project type.8 Organic Mode : a1 = 2.4, b1 = 1.05, c1 = 2.5, d1 = 0.38 Semidetached Mode : a1 = 3.0, b1 = 1.12, c1 = 2.5, d1 = 0.35 Embedded Mode : a1 = 3.6, b1 = 1.20, c1 = 2.5, d1 = 0.32			
23.	(a)	Identify the main components of a modern graphical user interface (GUI). Why are consistency and feedback important in GUI design? What are the essential steps involved designing user interface that optimizes usability and enhances user experiences?	8	3	3
		(OR)			
	(b)	i)Describe the key concepts of software design fundamentals and explain why they are important in the software development lifecycle. (3 Marks) ii)A banking system is a group or network of institutions that provide financial services for us. These institutions are responsible for operating a payment system, providing loans, taking deposits, Withdrawal of money and helping with investments. Develop a use case model for deposit and withdrawal process of banking system. (5 marks)			
24.	(a)	You are mentoring a group of junior developers who are unfamiliar with the concepts of desk checks, walkthroughs, and code reviews. Explain each practice and its importance in the software development process.	8	4	4
		(OR)			
	(b)	Evaluate the effectiveness of using white box testing techniques in a scenario where code coverage is critical. What are the potential risks and benefits of relying heavily on white box testing? Support your evaluation with examples.			
25.	(a)	As a software project manager in a company that specializes in the development of software for offshore oil industry, you have been given the task of discovering the factors that affect the maintainability of the systems developed by your company. How do you analyze the maintenance process models and discover appropriate maintenance process model for your company and explain the different types of the maintenances.	8	3	5
		(OR)			
	(b)	Define the key components of a risk management strategy. What are the main steps involved in identifying, assessing and mitigating risks in a project?			

PART - C (1 × 15 = 15 Marks)		**Marks**	**BL**	**CO**
Answer **any 1** Questions				
26.	You are a project manager overseeing the development of a new online banking application. The development team has implemented a login feature, which requires users to enter their username and password. The system should validate these credentials and grant access if they are correct. If the credentials are invalid, the system should display an error message.			

Your task is to create **manual test cases** to verify the functionality of the login feature based on the following requirements: 15 5 5

1. The user must enter a valid username and password to log in successfully.
2. If the username or password is incorrect, the system should display an appropriate error message.
3. The system should handle the scenario when either or both fields are left blank.
4. The system should lock the account after 3 failed login attempts, with a prompt indicating the account is temporarily locked.
5. Passwords should be encrypted and not visible in plain text.
6. After a successful login, the user should be redirected to the account dashboard.

Questions:

1. What is the importance of these manual test cases in software project management?
2. **Create at least 5 manual test cases to verify the login functionality,** ensuring you cover the mentioned requirements.
3. **For each test case, provide the following details:**
   - Test case ID
   - Description
   - Pre-conditions
   - Test Steps
   - Expected Result
   - Actual Result (to be filled after execution)
   - Status (Pass/Fail, to be filled after execution)

27. You are leading a software development project to build a new e-commerce platform for a client. The project is nearing its final stages, with the core functionality implemented and testing underway. However, you discover that the project is running behind schedule due to a critical bug requiring a significant amount of time and effort to fix. Simultaneously, you receive a request from the client to add a new feature (a loyalty program) that is not part of the original scope but is considered a high priority by the client. You have a limited team of developers and testers, and the project budget is fixed. The deadline is firm, and failing to deliver on time could result in a significant financial penalty. 15 3 2

    How would you approach this situation, considering the competing priorities of fixing the critical bug, potentially adding the new feature, and meeting the project deadline and budget constraints? What steps would you take to: Prioritize tasks effectively, Communicate the situation to the team and stakeholders, Manage the team's workload and resources, Resolve potential conflicts that might arise, and Ensure project success despite these challenges.

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